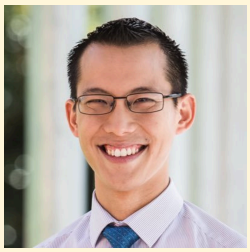
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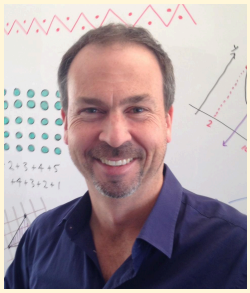
Ambassador Problem Picks



Eddie Woo shares The Two Rectangles problem



Kaye Stacey shares a Paper Folding Challenge



James Tanton shares the Finding Paths problem



Amie Albrecht shares the Connecting the Dots problem



Michaela Epstein shares the Spirolaterals challenge

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Eddie Woo's Favourite Problem: The Two Rectangles

Create two rectangles so that the first has a bigger perimeter, and the second has a bigger area.



Some questions to get you thinking:

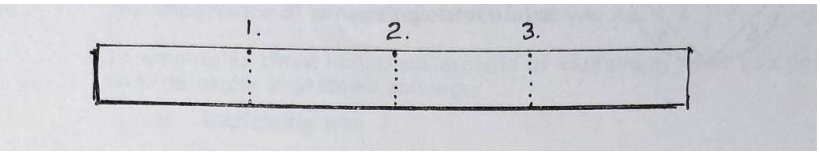
- Compare your rectangles with other students'. What do you notice?
- Find some more rectangle pairs. Can you find a shortcut?
- Can you create two rectangles so that the first has exactly double the perimeter of the second, and the second has exactly double the area of the first?

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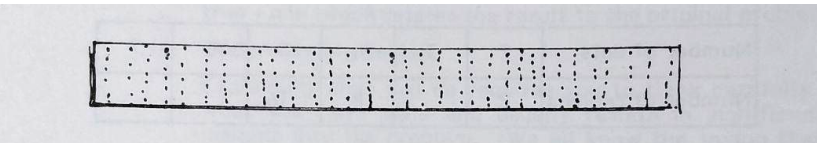
Kaye Stacey's Favourite Problem: The Paper Folding Challenge

Take a strip of paper about 30 cm long and 2 cm wide. Fold it in half, then half again. When you unfold it you will see that 3 creases have been made.



SOURCE: *Strategies for Problem Solving* (Stacey and Groves, 2006). Reproduced with permission of the author.

If you folded the strip in half **10 times**, how many creases would there be?



SOURCE: *Strategies for Problem Solving* (Stacey and Groves, 2006). Reproduced with permission of the author.

Some questions to get you thinking:

- How many folds is it possible to actually make?
- How might you keep track of what you're doing?
- Can you see a pattern? This can help with working systematically.
- What if you wanted to fold the strip in half 25 times? 100 times? *N* times?

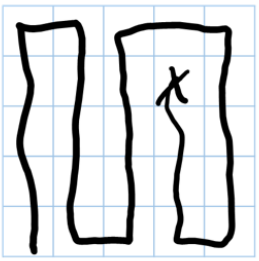
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James Tanton's Favourite Problem: Finding Paths

Pick a spot (any spot!) on a 5x5 grid . Can you walk a path (moving left, right, up or down) that follows each and every cell exactly once?

Here's a path that works:



Some questions to get you thinking:

- Are all starting points possible? (Are there any shortcuts you can take to work this out?)
- Are there any starting points where it's impossible to find a path?
- What if you didn't have a 5x5 grid, but had a...
  - 6x6 grid?
  - 5x6 grid?
  - 5x5x5 cube, where you visit each smaller cube?

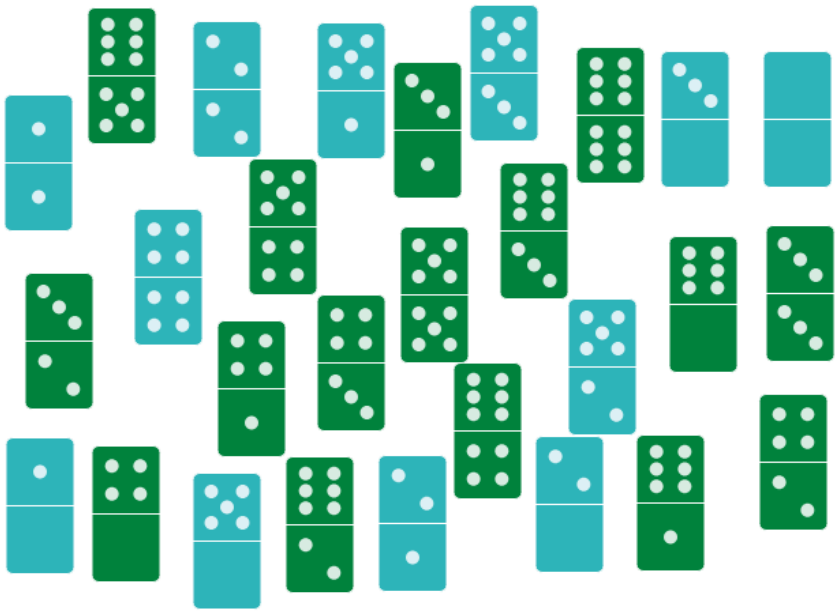
Learn more from James about this problem here.

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Amie Albrecht's Favourite Problem: Connecting the Dots

Is it possible to arrange an entire set of dominoes in a circle so that touching dominoes have adjacent squares with identical numbers?



Some questions to get you thinking:

- How many dominoes in a set?
- Is there exactly one of every combination?
- Can a domino have more than one instance of the same number?
- Can you find a solution for a dominoes set, in which the highest number is six?
- Is it possible to find a solution where the highest number is different?

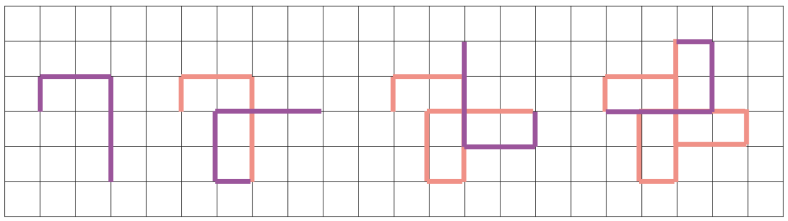
Learn more from Amie about this problem here.

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Michaela Epstein's Favourite Problem: Spirolaterals

Use three numbers to take a walk around a grid, and something interesting emerges.



This is a (1, 2, 3) Spirolateral. It's made by moving:

1. One space up,
2. Turning 90° clockwise. Moving 2 spaces to the right
3. Turning 90° clockwise. Moving 3 spaces down
4. Repeating steps 1-3 (turning 90° and moving 1, 2 or 3 spaces each time).

Pick any three numbers to make your own Spirolaterals. What do you notice? What shapes can you make?

Some questions to get you thinking:

- Can you make a spirolateral that looks like a square?
- Can you predict what shape you will make based on the 3 numbers you choose?
- What happens if you make a Spirolateral with 4 numbers? Or 5? 6? ...?
- What changes if you rotate by 60° instead of 90°? Or another angle?

Learn more from Michaela about this problem here.

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